Failure of Convexity under Dilation for the First Robin Eigenvalue

A connected-domain answer to a question of Keady and Wiwatanapataphee

Keio Research Notes

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Abstract

For the boundary condition $\beta\partial_\nu u + u = 0$, with $\beta > 0$ fixed, we construct bounded connected simply connected domains on which the first Robin eigenvalue is not convex as a function of the dilation factor. This gives a negative answer to the general-domain complete-monotonicity question posed by Keady and Wiwatanapataphee in 2018. The construction joins an 8×2 rectangle and a 3×3 square by a thin passage. The two chambers exchange which one has the smaller first eigenvalue. At three specified rational scales, exact rational arithmetic certifies a strict midpoint defect exceeding $1/16000$ for their minimum. A direct variational argument transfers this defect to every sufficiently thin positive-width passage. The same fixed planar domain works for every $\beta > 0$, and Cartesian products give counterexamples in every dimension at least two, also after unit-volume normalization. We further give an open parameter family in which reversal of the small- and large-scale ordering forces failure of complete monotonicity. All numerical-looking inequalities in the main construction are certified by finite rational calculations. The thin-passage technique is established; the contribution examined here is its application through chamber ground-state switching to the original dilation question.

1 The original question and the counterexample

For a bounded Lipschitz domain $D \subset \mathbb{R}^d$, write $\lambda_1(D;\beta)$ for the first eigenvalue of

$$-\Delta u = \lambda u \quad \text{in } D, \qquad \beta\partial_\nu u + u = 0 \quad \text{on } \partial D, \qquad \beta > 0. \tag{1}$$

The Robin parameter is constant on the entire exterior boundary. A smooth function $f : (0,\infty) \to \mathbb{R}$ is completely monotone if $(-1)^n f^{(n)} \geq 0$ for every integer $n \geq 0$. In particular, every such function is convex.

Keady and Wiwatanapataphee [1], Section 6, printed pages 924–925, ask whether the complete monotonicity of $s \mapsto \lambda_1(sD;0)$ persists for positive β, including when D is a rectangle. Their general setting on printed page 911 is a bounded simply connected domain with piecewise C^1 boundary; convexity of the domain is not assumed. The following theorem answers the arbitrary-domain assertion negatively in precisely that geometric class. It makes no claim of a counterexample among rectangles or convex domains.

Set

$$A = (-4,4) \times (-1,1), \qquad B = (6,9) \times (-3/2,3/2), \tag{2}$$

$$T_\varepsilon = (4,6) \times (-\varepsilon,\varepsilon), \qquad \Omega_\varepsilon = \operatorname{int}(\overline{A} \cup \overline{T}_\varepsilon \cup \overline{B}), \qquad 0 < \varepsilon < 1/4. \tag{3}$$

The interior-of-the-union convention includes the open attachment windows. Thus Ω_ε is a connected simply connected Lipschitz polygon, with area $25 + 4\varepsilon$.

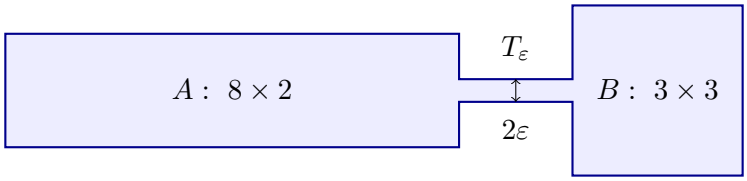


Figure 1: The explicit family of connected domains. The passage width is schematic: the theorem proves the result for all sufficiently small positive ε, without specifying a numerical threshold.

Theorem 1.1 (A strict failure of dilation convexity)**.** *Let*

$$s_- = \frac{1657}{1000}, \qquad s_0 = \frac{1667}{1000}, \qquad s_+ = \frac{1677}{1000}. \tag{4}$$

There exists $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\beta > 0$,

$$\lambda_1(\beta s_0 \Omega_\varepsilon; \beta) - \frac{\lambda_1(\beta s_- \Omega_\varepsilon; \beta) + \lambda_1(\beta s_+ \Omega_\varepsilon; \beta)}{2} > \frac{1}{32000\beta^2}. \tag{5}$$

Consequently, $s \mapsto \lambda_1(s\Omega_\varepsilon; \beta)$ is not convex and is not completely monotone. The domain can be chosen independently of β.

Theorem 1.2 (Every dimension and fixed volume)**.** *For every integer $d \geq 2$ there is a bounded connected simply connected Lipschitz polyhedral domain $D_d \subset \mathbb{R}^d$ such that $s \mapsto \lambda_1(sD_d; \beta)$ fails convexity for every $\beta > 0$. One may require $|D_d| = 1$.*

The proof has two distinct tasks. First we certify a strict finite-value inequality for a minimum of two explicit interval spectra. Then we realize that minimum as a limit of the first eigenvalues of connected domains. The latter step is a standard thin-passage idea, for which we give a complete variational proof. The former supplies the particular obstruction to the dilation property. Section 7 distinguishes this result from nearby literature and records the limits of the priority claim.

2 Interval coordinates and the scaling identities

It is useful to distinguish the reciprocal boundary conventions. For $\alpha > 0$, define

$$q_{D,\alpha}[u] = \int_D |\nabla u|^2 \, \mathrm{d}x + \alpha \int_{\partial D} |\operatorname{Tr} u|^2 \, \mathrm{d}\sigma, \qquad u \in H^1(D), \tag{6}$$

$$\Lambda(D; \alpha) = \inf_{u \neq 0} \frac{q_{D,\alpha}[u]}{\|u\|^2_{L^2(D)}}. \tag{7}$$

Thus $\Lambda(D; \alpha)$ uses $\partial_\nu u + \alpha u = 0$, and $\lambda_1(D; \beta) = \Lambda(D; 1/\beta)$. Changing variables in the Rayleigh quotient gives

$$\lambda_1(sD; \beta) = s^{-2}\Lambda(D; s/\beta), \qquad \lambda_1(tD; \beta) = \beta^{-2}\lambda_1((t/\beta)D; 1). \tag{8}$$

In particular, the parameter transfer in Theorem 1.1 is exact and involves no uniform-in-β limiting argument.

For $y > 0$, let $\mu(y) > 0$ be determined by

$$y\mu(y) = \arctan \frac{1}{\mu(y)}, \quad \text{equivalently} \quad H_y(\mu(y)) = 0, \qquad H_y(x) = yx + \arctan x - \frac{\pi}{2}. \tag{9}$$

Since $H_y'(x) = y + (1 + x^2)^{-1} > 0$, there is a unique positive solution. The implicit function theorem shows that μ is real analytic. Also

$$\mu'(y) = -\frac{\mu(y)}{y + (1 + \mu(y)^2)^{-1}} < 0, \qquad 0 < \mu(y) < \frac{\pi}{2y}. \tag{10}$$

Lemma 2.1 (The rectangle formula)**.** *For a rectangle with half-side lengths $a, b > 0$,*

$$\lambda_1(s((-a,a)\times(-b,b));1) = \mu(as)^2 + \mu(bs)^2. \tag{11}$$

Translations of the rectangle do not change this formula.

Proof. On the interval $(-y, y)$, the positive even ground state is $\cos(\mu(y)x)$; the boundary condition is $\mu(y)\tan(y\mu(y)) = 1$ with $0 < y\mu(y) < \pi/2$. This is equivalent to (9). The one-dimensional variational characterization gives a lower bound by $\mu(y)^2$ for each slice. Apply that bound separately to the two coordinate directions and integrate. The product of the two cosine ground states attains the resulting lower bound, proving the formula. □

Applied to (2), the formula gives two analytic branches and their minimum:

$$a(s) = \mu(s)^2 + \mu(4s)^2, \qquad b(s) = 2\mu(3s/2)^2, \qquad F(s) = \min\{a(s), b(s)\}. \tag{12}$$

These are exactly the first eigenvalues of the two scaled chambers and their disconnected union. The union is only an intermediate model; the final examples have a positive-width passage.

3 Designing an order reversal

The auxiliary design problem is to make two elementary chamber spectra exchange their order as the scale changes. This can be done for every nonsquare rectangle.

Lemma 3.1 (Endpoint asymptotics)**.** *The interval function satisfies*

$$\mu(y)^2 \sim \frac{1}{y} \quad (y \downarrow 0), \qquad \mu(y)^2 \sim \frac{\pi^2}{4y^2} \quad (y \to \infty). \tag{13}$$

Proof. The function $x \mapsto \arctan(1/x)/x$ decreases continuously from $+\infty$ to 0. Hence its inverse tends to $+\infty$ as $y \downarrow 0$, and to 0 as $y \to \infty$. In the first limit, (9) gives $y\mu(y)^2 = \mu(y)\arctan(1/\mu(y)) \to 1$. In the second it gives $y\mu(y) \to \pi/2$. □

Proposition 3.2 (An open family of competing chambers)**.** *Let $a, b > 0$ with $a \neq b$, and choose c in the nonempty interval*

$$\frac{\sqrt{2}\,ab}{\sqrt{a^2+b^2}} < c < \frac{2ab}{a+b}. \tag{14}$$

Set

$$R(s) = \mu(as)^2 + \mu(bs)^2, \qquad Q(s) = 2\mu(cs)^2.$$

Then $R(s) < Q(s)$ for all sufficiently small $s > 0$, while $R(s) > Q(s)$ for all sufficiently large s. The function $\min\{R, Q\}$ is not real analytic on $(0,\infty)$ and is not completely monotone.

Proof. The asymptotic coefficients in Lemma 3.1 are

$$R(s) \sim \frac{a^{-1}+b^{-1}}{s}, \qquad Q(s) \sim \frac{2}{cs} \qquad (s \downarrow 0),$$
$$R(s) \sim \frac{\pi^2}{4s^2}(a^{-2}+b^{-2}), \qquad Q(s) \sim \frac{\pi^2}{4s^2}\frac{2}{c^2} \qquad (s \to \infty).$$

The two strict inequalities in (14) are precisely the inequalities that reverse these orderings. The interval is nonempty because $2(a^2+b^2) > (a+b)^2$ when $a \neq b$.

If the minimum were real analytic, its agreement with R on a nonempty open interval would, by the real-analytic identity theorem, force agreement on all of $(0,\infty)$. This contradicts the opposite strict ordering at large scales. Complete monotonicity implies real analyticity by the Laplace-transform representation, so it is excluded as well; see [6]. □

For $(a, b, c) = (4, 1, 3/2)$, the comparison reduces to

$$\frac{5}{4} < \frac{4}{3}, \qquad \frac{17}{64} > \frac{2}{9}. \tag{15}$$

The small-scale coefficients are the perimeter-to-area ratios of the two chambers. The large-scale coefficients are their first Dirichlet eigenvalues divided by π^2.

Order reversal alone gives a qualitative obstruction to complete monotonicity. It does not by itself prove failure of convexity: two analytic branches might meet to higher order. The next section supplies a strict midpoint witness, which is the only finite calculation used in the stronger theorem.

4 A rational certificate for the midpoint defect

Proposition 4.1 (Exact finite witness)**.** *For F in (12) and the scales in (4),*

$$F(s_0) - \frac{F(s_-) + F(s_+)}{2} > \frac{674670894519}{10^{16}} > \frac{1}{16000}. \tag{16}$$

Proof. We give finite rational bounds for the six interval roots that are needed. For $0 < x < 1$, define

$$S_m(x) = \sum_{k=0}^{m} \frac{(-1)^k x^{2k+1}}{2k+1}.$$

The alternating-series estimate gives

$$S_{2n+1}(x) < \arctan x < S_{2n}(x). \tag{17}$$

Using Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, put

$$\begin{aligned} p_- &= 16S_{21}(1/5) - 4S_{20}(1/239), \\ p_+ &= 16S_{20}(1/5) - 4S_{21}(1/239). \end{aligned} \tag{18}$$

Then $p_- < \pi < p_+$. For completeness, Machin's identity follows by applying the tangent addition formula twice: $\tan(4\arctan(1/5)) = 120/119$, and subtracting $\arctan(1/239)$ gives tangent 1. The angle lies in $(0, \pi/2)$, so it is $\pi/4$.

In Table 1, set $\ell_j = n_j/10^8$ and $u_j = (n_j+1)/10^8$. The last two columns are integer certificates for the following strictly rational inequalities:

$$10^9(y_j\ell_j + S_{40}(\ell_j) - p_-/2) < U_j < 0, \tag{19}$$

$$10^9(y_j u_j + S_{41}(u_j) - p_+/2) > V_j > 0. \tag{20}$$

They are obtained by substituting the displayed integers into the finite sums; a complete standard-library rational verifier is included in Appendix A. By (17)–(18), these inequalities imply $H_{y_j}(\ell_j) < 0 < H_{y_j}(u_j)$. Strict increase of H_{y_j} therefore proves

$$\ell_j < \mu(y_j) < u_j \qquad (1 \le j \le 6). \tag{21}$$

At the center, both branches require lower bounds. At the endpoints, it suffices to bound any one branch from above:

$$\begin{aligned} F(s_0) &> \min\{\ell_3^2 + \ell_4^2,\, 2\ell_5^2\}, \\ F(s_-) &< u_1^2 + u_2^2, \qquad F(s_+) < 2u_6^2. \end{aligned}$$

Table 1: Exact brackets of width 10^{-8} for the six roots. All four numerical columns contain exact rational or integer data.

j	Role	y_j	n_j	U_j	V_j
1	$\mu(s_-)$	1657/1000	61515335	-2	21
2	$\mu(4s_-)$	6628/1000	20629921	-72	3
3	$\mu(s_0)$	1667/1000	61258304	-1	21
4	$\mu(4s_0)$	6668/1000	20521733	-17	58
5	$\mu(3s_0/2)$	5001/2000	45682209	-16	17
6	$\mu(3s_+/2)$	5031/2000	45477263	-23	9

There is no assumption about which branch attains the minimum at either endpoint. Direct integer arithmetic gives

$$\begin{aligned}\min\{\ell_3^2+\ell_4^2,\,2\ell_5^2\}-\tfrac{1}{2}(u_1^2+u_2^2+2u_6^2)\\ =\frac{674670894519}{10^{16}}=\frac{1}{16000}+\frac{49670894519}{10^{16}}.\end{aligned}$$

Both strict comparisons in (16) follow. □

The certificate uses no decimal approximation as a premise. The alternating sums, the Machin bounds, the root enclosures, and the final comparison are all finite rational statements. In particular, no numerical crossing location or derivative estimate is required.

5 An elementary limit for a thin passage

Lemma 5.1 (Convergence to the chamber minimum)**.** *For every fixed $\alpha>0$,*

$$\lim_{\varepsilon\downarrow 0}\Lambda(\Omega_\varepsilon;\alpha)=\min\{\Lambda(A;\alpha),\Lambda(B;\alpha)\}. \tag{22}$$

The same statement holds for two fixed rectangles attached at the centers of facing sides by a straight passage of any fixed positive length and width 2ε.

Proof. We prove both inequalities directly. Compactness is used only on the two fixed chambers.

Upper estimate. Choose a chamber $D\in\{A,B\}$ and its normalized first Robin eigenfunction v for the coefficient α. Separation of variables makes v a product of cosine functions, so v and ∇v are bounded on the closed rectangle. Let P be the center of the attachment window. Put

$$r_\varepsilon=2\varepsilon,\qquad R_\varepsilon=\sqrt{\varepsilon},\qquad L_\varepsilon=\log(R_\varepsilon/r_\varepsilon).$$

For sufficiently small ε, define

$$\chi_\varepsilon(x)=\begin{cases}0, & |x-P|\le r_\varepsilon,\\ \log(|x-P|/r_\varepsilon)/L_\varepsilon, & r_\varepsilon<|x-P|<R_\varepsilon,\\ 1, & |x-P|\ge R_\varepsilon.\end{cases}$$

Its exceptional region has area at most $\pi\varepsilon$, and polar integration gives

$$\int_{\mathbb{R}^2}|\nabla\chi_\varepsilon|^2\,\mathrm{d}x=\frac{2\pi}{L_\varepsilon}\longrightarrow 0. \tag{23}$$

By the product rule and boundedness of $v,\nabla v$, this proves $\chi_\varepsilon v\to v$ strongly in $H^1(D)$. The trace operator on the fixed rectangle is continuous, so the boundary traces converge in $L^2(\partial D)$.

Extend $\chi_\varepsilon v$ by zero to the rest of Ω_ε. This extension belongs to $H^1(\Omega_\varepsilon)$ because it vanishes on a neighborhood of the entire window. The Rayleigh quotient on the joined domain tends to $\Lambda(D;\alpha)$. Choosing either chamber gives

$$\limsup_{\varepsilon\downarrow 0} \Lambda(\Omega_\varepsilon;\alpha) \le \min\{\Lambda(A;\alpha),\Lambda(B;\alpha)\}. \tag{24}$$

The passage cannot retain mass at bounded energy. For $f \in H^1(-\varepsilon,\varepsilon)$, the fundamental theorem of calculus and Cauchy–Schwarz give

$$\int_{-\varepsilon}^{\varepsilon} |f(y)|^2\,\mathrm{d}y \le 4\varepsilon|f(\varepsilon)|^2 + 8\varepsilon^2 \int_{-\varepsilon}^{\varepsilon} |f'(y)|^2\,\mathrm{d}y.$$

Apply this to transverse slices of $u \in H^1(\Omega_\varepsilon)$ and integrate along the passage. Its top side is part of the exterior Robin boundary. Thus

$$\int_{T_\varepsilon} |u|^2\,\mathrm{d}x \le \left(\frac{4\varepsilon}{\alpha} + 8\varepsilon^2\right) q_{\Omega_\varepsilon,\alpha}[u]. \tag{25}$$

The positive Robin coefficient on the lateral boundary is essential in this estimate.

Lower estimate. Choose a sequence $\varepsilon \downarrow 0$ realizing the lower limit of the eigenvalues, and normalized first eigenfunctions u_ε. The upper estimate bounds their energies. Their restrictions to A and B are bounded in the two fixed H^1 spaces. After passing to a subsequence, these restrictions converge weakly in H^1 and strongly in L^2 to u_A, u_B. Equation (25) implies

$$\|u_A\|^2_{L^2(A)} + \|u_B\|^2_{L^2(B)} = 1. \tag{26}$$

For $\delta > 0$, omit from each chamber boundary the part in the radius-δ disk about its attachment point P_D. For all sufficiently small ε, the remaining fixed boundary portions belong to $\partial\Omega_\varepsilon$. Weak continuity of the trace on each fixed chamber and weak lower semicontinuity yield

$$\liminf_{\varepsilon\downarrow 0} q_{\Omega_\varepsilon,\alpha}[u_\varepsilon] \ge \sum_{D\in\{A,B\}} \left(\int_D |\nabla u_D|^2\,\mathrm{d}x + \alpha \int_{\partial D\setminus B(P_D,\delta)} |\operatorname{Tr} u_D|^2\,\mathrm{d}\sigma\right).$$

Let $\delta \downarrow 0$. The limiting omitted sets are single points of zero boundary measure. Monotone convergence, the Rayleigh principle, and (26) give

$$\begin{aligned}\liminf_{\varepsilon\downarrow 0} \Lambda(\Omega_\varepsilon;\alpha) &\ge q_{A,\alpha}[u_A] + q_{B,\alpha}[u_B] \\ &\ge \min\{\Lambda(A;\alpha),\Lambda(B;\alpha)\}.\end{aligned}$$

Together with (24), this proves the limit. The argument depends only on fixed rectangular chambers, side-center attachments, and a fixed positive passage length, proving the last assertion as well. □

6 Transfer to connected domains and all dimensions

Proof of Theorem 1.1. By (8), Lemma 5.1, and Lemma 2.1, for every fixed $s > 0$,

$$f_\varepsilon(s) := \lambda_1(s\Omega_\varepsilon;1) \longrightarrow F(s). \tag{27}$$

Let $\eta = 1/16000$. Proposition 4.1 gives a limiting midpoint defect greater than η. Convergence at the three fixed scales permits the simultaneous estimates

$$|f_\varepsilon(s_j) - F(s_j)| < \eta/4, \qquad j \in \{-,0,+\},$$

for all $0 < \varepsilon < \varepsilon_0$ after choosing one $\varepsilon_0 > 0$. The total error in the midpoint defect is less than $\eta/2$. Consequently

$$f_\varepsilon(s_0) - \frac{f_\varepsilon(s_-) + f_\varepsilon(s_+)}{2} > \eta/2 = \frac{1}{32000}. \tag{28}$$

This violates the defining midpoint inequality for convexity. Finally, the second identity in (8) multiplies this gap by β^{-2} and the three scales by β, giving (5). No convergence of derivatives, unidentified eigenvalue crossing, or conjectural regularity is used. □

The assertion "for all sufficiently small positive ε" is an unconditional existence statement about actual connected polygons. The proof does not specify a numerical value of ε_0, and it does not certify a selected decimal passage width. Such an explicit threshold would require a quantitative refinement of Lemma 5.1, but is not needed for the stated counterexample.

Proof of Theorem 1.2. Fix any ε satisfying Theorem 1.1. The case $d = 2$ is already proved. For $d \geq 3$, set

$$L = 1000d, \qquad D_{d,\varepsilon} = \Omega_\varepsilon \times (-L, L)^{d-2}.$$

This is a bounded connected simply connected Lipschitz polyhedral domain. The first eigenvalue on a Cartesian product is the sum of the first eigenvalues on its factors: the lower bound follows by applying the variational inequality on slices, and the product of ground states attains it. Hence

$$\lambda_1(sD_{d,\varepsilon}; 1) = f_\varepsilon(s) + (d-2)\mu(sL)^2.$$

Since μ is decreasing, the added term has midpoint defect at least $-(d-2)\mu(s_-L)^2$. From (10), $s_- > 1$, and $\pi < 4$,

$$(d-2)\mu(s_-L)^2 < \frac{(d-2)\pi^2}{4s_-^2L^2} < \frac{4}{10^6} < \frac{1}{64000}. \tag{29}$$

Combining with (28) leaves a midpoint defect greater than $1/64000$. The exact parameter transfer in (8) proves the assertion for every $\beta > 0$.

For unit volume, put

$$V_{d,\varepsilon} = (25 + 4\varepsilon)(2L)^{d-2}, \qquad \widetilde{D}_{d,\varepsilon} = V_{d,\varepsilon}^{-1/d} D_{d,\varepsilon}.$$

Then the violating scales are $t_j = \beta V_{d,\varepsilon}^{1/d} s_j$, because $t_j \widetilde{D}_{d,\varepsilon} = \beta s_j D_{d,\varepsilon}$. For $d = 2$, use $V_{2,\varepsilon} = 25+4\varepsilon$ and the same identity. This proves the volume-normalized assertion. □

Corollary 6.1 (The qualitative parameter family)**.** *Let a, b, c satisfy (14). Translate a rectangle of half-sides a, b and a square of half-side c so that two side centers face each other across a gap of fixed positive length. Join them by a straight passage of width 2ε. For all sufficiently small positive ε, the first Robin eigenvalue on the joined domain is not completely monotone under dilation, for every $\beta > 0$.*

Proof. Lemma 5.1 gives pointwise convergence of the dilation curves to $\min\{R, Q\}$ from Proposition 3.2. Completely monotone functions are closed under pointwise limits that are finite on $(0, \infty)$; this follows from their Laplace-transform representation [6]. If completely monotone examples existed at arbitrarily small positive widths, one could select such a sequence $\varepsilon_j \downarrow 0$. The finite pointwise limit would then be completely monotone, contradicting Proposition 3.2. This proves the asserted "all sufficiently small" quantifier. Transfer from $\beta = 1$ to every positive β is again exact. □

The nonconvexity established in Theorems 1.1 and 1.2 also excludes log-convexity and a positive Laplace-transform representation for those dilation curves. These are consequences of the same midpoint witness, not independent solutions of additional problems. At $\beta = 0$, the usual Dirichlet dilation formula remains $\lambda_1(sD; 0) = s^{-2}\lambda_1(D; 0)$, which is completely monotone. There is no contradiction as $\beta \downarrow 0$: the violating scales in (5) simultaneously tend to zero.

7 Relation to previous work and scope of the claim

The target is the existing question in Section 6 of [1], not a new problem obtained by changing its boundary convention, connectedness assumption, or class of domains. The original paper proves dilation convexity for boxes in Theorem 3(i). Our domain is a simply connected nonconvex polygon, as permitted in the general setting. The rectangle-restricted question is distinct from the arbitrary-domain assertion refuted here.

The exact interval formula is also established in [1]. Its inverse-arctangent viewpoint is developed in [2]. The present proof uses only that formula, elementary endpoint asymptotics, and finite rational bounds; it does not assume any earlier research claim that the interval inverse is completely monotone. Thus the negative result is logically independent of an affirmative resolution for boxes.

Thin-passage spectral approximation and logarithmic capacity cutoffs are established techniques. In particular, Dai and Sun [3], Lemma 4.2 and Proposition 4.3, use small Robin windows and thin passages in their counterexamples to Laugesen's eigenvalue-ratio conjecture [4]. Their quantity is λ_2/λ_1 as a function of the boundary coefficient; their unequal cubes retain a fixed ordering of their component ground states. The present quantity is the first eigenvalue under domain dilation at a fixed Robin parameter. Here the rectangle and square exchange their ground-state ordering. We credit the earlier geometric machinery and do not claim to have invented it.

The reusable part of the construction is to arrange opposite endpoint orderings, compute the lower envelope, and transfer a strict finite-value defect through a spectral limit. The open interval (14) supplies a systematic design rule; the exact certificate strengthens the qualitative obstruction to an ordinary convexity violation. Neither taking a minimum nor a stability argument for strict inequalities is claimed as a historically new abstract principle.

A literature check through 31 October 2026 covered the original papers, the Robin spectral-conjecture paper [4], the recent thin-passage work [3], the scaling work [5], and all 95 TeX manuscripts in the supplied corpus snapshot [7]. No identical dilation-convexity counterexample was located in the material checked. This is a bounded search finding, not a proof that no unindexed or unpublished antecedent exists. The original item is an attributable published open question; we do not assert that it was a widely established named conjecture.

The mathematical conclusions of the theorems are unconditional. Publication priority, external refereeing, and comparative impact are separate questions. In particular, no claim of superiority to the highest-ranked results in the supplied corpus follows from the proof here.

A A complete rational verifier

The following Python 3 program uses only the standard-library `fractions` module. It checks every sign certificate in Table 1, the identity for the rational midpoint lower bound, and its strict comparison with 1/16000. Numerical root finding is not part of the verification. The source is included here so the single TeX file contains the entire finite certificate.

```
from fractions import Fraction as Q

def S(x, m):
    return sum(((-1)**k * x**(2*k+1) / (2*k+1)
                for k in range(m+1)), Q(0))

p_lo = 16*S(Q(1,5),21) - 4*S(Q(1,239),20)
p_hi = 16*S(Q(1,5),20) - 4*S(Q(1,239),21)
assert p_lo < p_hi

# y, lower-bracket numerator, upper sign, lower sign
rows = [
```

```
    (Q(1657,1000), 61515335, -2, 21),
    (Q(6628,1000), 20629921, -72, 3),
    (Q(1667,1000), 61258304, -1, 21),
    (Q(6668,1000), 20521733, -17, 58),
    (Q(5001,2000), 45682209, -16, 17),
    (Q(5031,2000), 45477263, -23, 9),
]
bounds = []
for y, n, U, V in rows:
    lo, hi = Q(n,10**8), Q(n+1,10**8)
    assert 0 < lo < hi < 1
    assert 10**9*(y*lo + S(lo,40) - p_lo/2) < U < 0
    assert 10**9*(y*hi + S(hi,41) - p_hi/2) > V > 0
    bounds.append((lo, hi))

l1,u1 = bounds[0]
l2,u2 = bounds[1]
l3,u3 = bounds[2]
l4,u4 = bounds[3]
l5,u5 = bounds[4]
l6,u6 = bounds[5]
gap = min(l3*l3 + l4*l4, 2*l5*l5) - (u1*u1 + u2*u2 + 2*u6*u6)/2
assert gap == Q(674670894519,10**16)
assert gap - Q(1,16000) == Q(49670894519,10**16)
assert gap > Q(1,16000)
print("All exact inequalities verified.")
```

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